kunalmittal459@gmail.com | +91 6280003218

CPI Inflation Analysis (India, 2013-2023)

Excel | Data Cleaning, Imputation, Formulas, PivotTables, Charts | 372 monthly records

1. PROBLEM

Consumer Price Index (CPI) data shows how prices change, but a single number hides the detail. The goal was to answer five questions: which category contributes most to the CPI, which year had the highest year-on-year inflation, how food prices moved month to month, how COVID-19 affected inflation (food, health and essentials), and how imported oil prices relate to inflation in every category.

2. DATA

372 monthly records for three sectors (Rural, Urban, Rural+Urban) from January 2013 to May 2023: category-wise CPI indices (cereals, meat and fish, egg, milk, oils and fats, fruits, vegetables, pulses, sugar, spices, beverages, prepared meals, clothing, footwear, housing, fuel and light, household goods, health, transport and communication, recreation, education, personal care, pan and tobacco, miscellaneous) and the General Index. Categories were grouped into 12 broader buckets (Food, Clothing and Footwear, Housing, Energy, Household, Health, and others) using a mapping sheet. For objective 5, imported oil prices (Jan 2021 - May 2023) were collected from different websites and joined to the CPI data.

3. DATA CLEANING: PROBLEMS FOUND AND WHAT I DID

Problem found	Action taken
Month "Marcrh" misspelled	Corrected to March
Possible duplicates	Checked sector + month + year: no duplicates
Cereals: 3 missing values (May 2020, one per sector); Egg: same pattern	Filled using a 2-point moving average of neighbouring months
Meat and fish: 6 missing values (April and May 2020, three sectors)	Filled using linear interpolation (constant rate of change between known values)
Other columns (milk to General Index) and Housing in May 2020	Linear interpolation or 2-point moving average, as applicable
Housing (Rural): missing or non-numeric values throughout	Not imputed: too few neighbouring values for a reliable estimate; documented
April 2019 absent for all three sectors in the source data	Left out and documented, not estimated

4. METHOD FOR EACH OBJECTIVE

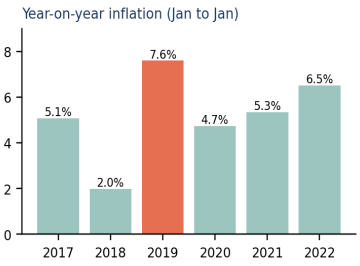
Objective	How it was done
1. Category contribution	Latest month (May 2023); each broader bucket index divided by the sum of all bucket indices (share of total, sums to 100%)
2. Year-on-year inflation	General Index, January to the next January (Rural+Urban): (end index - start index) / start index; 2023 excluded (no full 12 months)
3. Food trends	Broader food bucket index for May 2022 - May 2023; month-on-month % change; 12-month % change for each food category (Jun 2022 vs May 2023)
4. COVID-19	Dec 2019 - Dec 2020: month-on-month % for General Index, Food, Health and an Essential bucket, with periods before, at onset of and after the first lockdown
5. Imported oil	Oil price month-on-month % (Jan 2021 - May 2023) vs month-on-month % of the General Index and each category; correlation (CORREL) for each

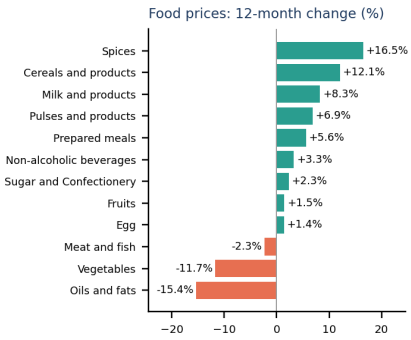
5. RESULTS (MATCH THE OBJECTIVE SHEETS IN THE EXCEL FILE)

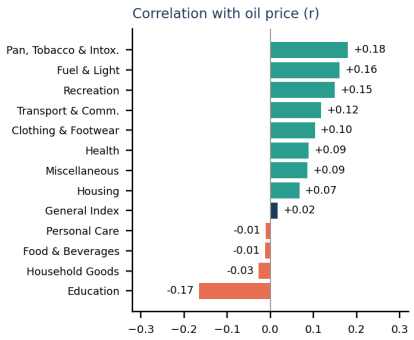
Objective	Result
1. Category contribution	Share of the index total in May 2023: Pan, Tobacco and Intoxicants highest (9.3%), then Health (8.6%) and Personal Care (8.6%); Transport and Communication lowest (7.6%). Shares are close because every bucket index is high; this is an index-share view, not a weighted contribution.
2. Year-on-year inflation	**2019 (Jan 2019 to Jan 2020): 7.6%**, the highest (index 139.6 to 150.2); lowest 2018: 2.0%. Others: 2017 5.1%, 2020 4.7%, 2021 5.3%, 2022 6.5%.
3. Food prices	Food index 172.1 (May 2022) to 177.3 (May 2023). Month-on-month: highest +1.04% (Jun 2022), lowest -0.66% (Feb 2023). 12-month change: spices +16.5% and cereals +12.1% rose most; oils and fats -15.4%, vegetables -11.7% fell.
4. COVID-19	General Index fell from 150.4 (Dec 2019) to 148.6 (Mar 2020 lockdown), then rose to 158.9 (Dec 2020). Average monthly inflation: -0.43% before, -0.34% at onset, +0.75% after. Food +3.5% in Apr 2020 and +8.5% over 2020; Health up in 10 of 12 months (about +0.4% a month); Essentials +3.3% over 2020.
5. Imported oil	Oil rose 133% (Dec 2020 to Jun 2022) while the General Index rose 8.6%; oil then fell 35% to May 2023 while the index still rose 3.8%. Correlation with the General Index 0.02; with categories between -0.17 (Education) and +0.18 (Pan, Tobacco): all weak.

CPI Inflation Analysis (continued)

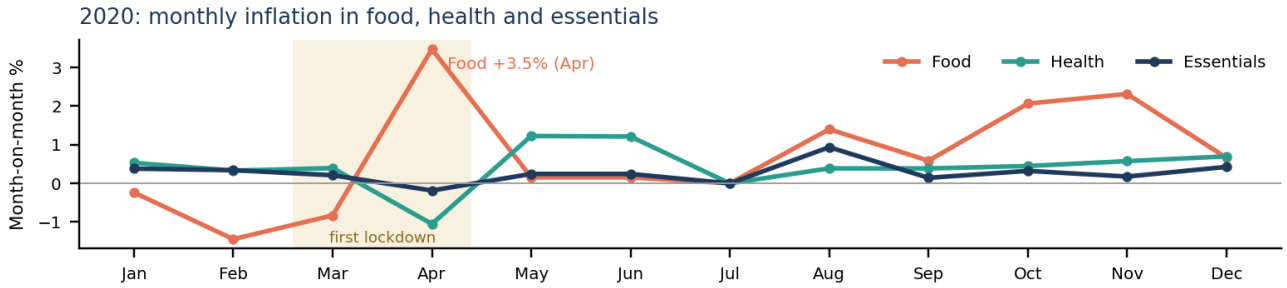
6. CHARTS







Correlation chart: r runs from -1 to +1; values near 0 mean almost no link between month-to-month oil price changes and price changes in that category.



7. CONCLUSIONS

- Inflation rose over the decade and peaked in the 12 months before COVID (7.6%).
- Food prices moved unevenly: cereals and spices rose while oils and vegetables fell, so one headline number hides the story.
- Early in the 2020 lockdown, food prices jumped while essentials stayed steady; health prices rose almost every month.
- Imported oil price swings did not explain inflation in 2021-2023: correlations were weak for every category.

8. LIMITATIONS

Interpolated and moving-average values are estimates (April-May 2020 especially), so swings in those months need care. April 2019 is missing from the source, and Rural Housing could not be imputed. The category share in objective 1 does not use CPI weights. Correlation does not prove cause. The oil series was compiled from several websites and is used as collected.

SKILLS SHOWN

Data cleaning and imputation | Year-on-year and month-on-month analysis | Correlation | PivotTables and charts | Data sourcing and documentation